# The Next Generation of Measurements



# Introduction

The Measurement Revolution driving today's innovation in test and measurement was born through consistent – if not constant – PC technology innovation. With innovation comes change. Evolution in measurement systems is nothing new; in fact, the way engineers, scientists, and technicians take measurements has perpetually evolved, albeit at a slower pace than the PC or telecom industries. Digital technology served as the catalyst that transformed measurement systems from the early analog instruments to today's modern digital instruments. PC technologies made virtual instrumentation a reality. However, the PC has not been the only driving force in creating Digital Measurement Systems – the next evolution in virtual instrumentation. Innovations in software, communications technologies, and consumer electronics have also been instrumental in the profound changes occurring in the test and measurement industry today – the Measurement Revolution. This paper explores the evolution in measurement components that has made Digital Measurement Systems possible and presents a framework for building these powerful systems.

# Analog Instruments

Just like the first electronic devices, early instruments were built with discrete analog and mechanical components. For example, early meters used a coil and a needle. The needle traveled back and forth across a scale on the instrument face to indicate a measured reading in proportion to the magnitude of applied stimulus. To take a measurement, you simply read the position where the needle settled. Analog oscilloscopes, the first oscilloscopes, refracted a beam of charged particles toward a phosphor-coated screen to reflect the frequency and amplitude of the measured signal on a grid-line display. From a functional perspective, analog instruments follow the diagram in Figure 1, where the actual acquisition of the signal merges with the presentation so the measured value can be read directly from the display.

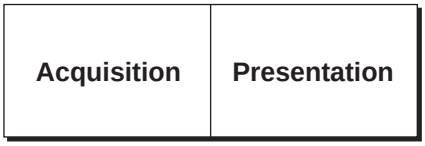



**Figure 1.** Building Blocks of Analog Instrumentation

Analog instruments evolved into instruments that use digital technology, or digital instruments. Instead of discrete analog and mechanical components, digital instruments use analog-to-digital converters (ADCs) to digitize analog signals into a numerical representation. Once data is acquired, the value(s) are stored into memory located inside the instrument. From there, the data can be presented on the instrument display – just like analog instruments – or further processed or analyzed to expand the measurement capability and utility of the instrument.

# Introducing Analysis to Instruments

With the introduction of digital technology to instruments, a significant step for the industry, instrument vendors could embed intelligence or microprocessor technology directly inside their instruments. By integrating "intelligence," instrument vendors created new features that took advantage of the embedded microprocessor with custom software. Adding signal processing, mathematical operations, communication interfacing, and/or configuration enhancements such as "autosetup" involved developing the software or firmware deployed inside the instrument. Consequently, stand-alone instrument vendors transformed the instrument paradigm from Figure 1 to Figure 2 by decoupling the acquisition from the presentation and adding analysis capability in between. Stand-alone digital instruments or digitizers essentially became computers with built-in ADCs.




**Figure 2.** Modern Instrument Architecture

To further illustrate this point, Figure 3 shows the block diagram of the components of a stand-alone digital instrument. Notice how the internal architecture of the stand-alone instrument resembles the architecture of a PC with a measurement board installed in an expansion slot. In both systems, the microprocessor or CPU controls the operation of the instrument and the ADC digitizes the signal. The communications interface connects the instrument to the PC with a stand-alone instrument. However, in a computer-based measurement system, the communications interface is the expansion bus. Today's computers typically use PCI as the standard expansion interface. The two paradigms fundamentally differ in the software deployed and the measurement capabilities of the digitizing hardware or ADC.

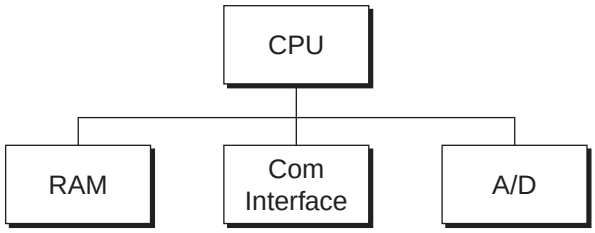



**Figure 3.** Block Diagram of Stand-Alone Digital Instrument

National Instruments pioneered virtual instrumentation in the 1980s by advocating that acquisition, analysis, and presentation need not reside in the same box. In a virtual instrumentation system, these elements can be distributed between the instrument (stand-alone, modular, or PC plug-in board) and the system computer. Furthermore, the advances in networking technology and the vast investments in telecommunications and the Internet make it easy to distribute these elements across several computers.

# The Ever Evolving ADC

The proliferation of digital instruments sparked the renaissance of the ADC. Today, there are several types of ADCs that vary widely in resolution, sampling rate, and speed (see Table A). Instruments deploy the ADC best suited to their designated function and the type of measurements defined by the manufacturer. DMMs, for example, may employ an "integrating ADC," which typically has good accuracy, but takes measurements slowly. Flash converters feature high sampling rates, but typically lower resolution than integrating ADCs. Successive approximation ADCs offer higher accuracy than flash converters with sampling rates landing somewhere in between integrating ADCs and flash converters. Delta-sigma ADCs use oversampling techniques combined with antialiasing and decimation filters to produce highly accurate measurements particularly honed for frequency analysis applications such measuring audio, sound, and vibration signals.

**Table 1.** ADC Comparison

| Type of ADC | Typical Application | Sampling Rate | Resolution |
|---|---|---|---|
| Successive Approximation | Data Acquisition | 5 MHz | < 16 bits |
| Integrating | DMMs | 1 kHz | > 20 bits |
| Flash | Oscilloscopes | 1 GHz | 8 to 10 bits |
| Delta-Sigma | Frequency Analyzers | 192 kHz | < 18 bits |

Since the birth of digital instruments, stand-alone instrument manufacturers have invested resources in developing ADC technology to offer products with unique and differentiated measurement capabilities. In contrast, computer-based instrument and data acquisition board manufacturers developing PXI/CompactPCI modules and/or PCI plug-in boards typically use off-the-shelf ADC technologies. Figure 4 plots the ADC technologies used by both types of instruments. In the past, there was a large gap in ADC technology between the two; however, technology innovations continually narrow this gap. Currently, the biggest differences appear at the extremes of the graph where cost effective off-the-shelf ADCs have yet to emerge.

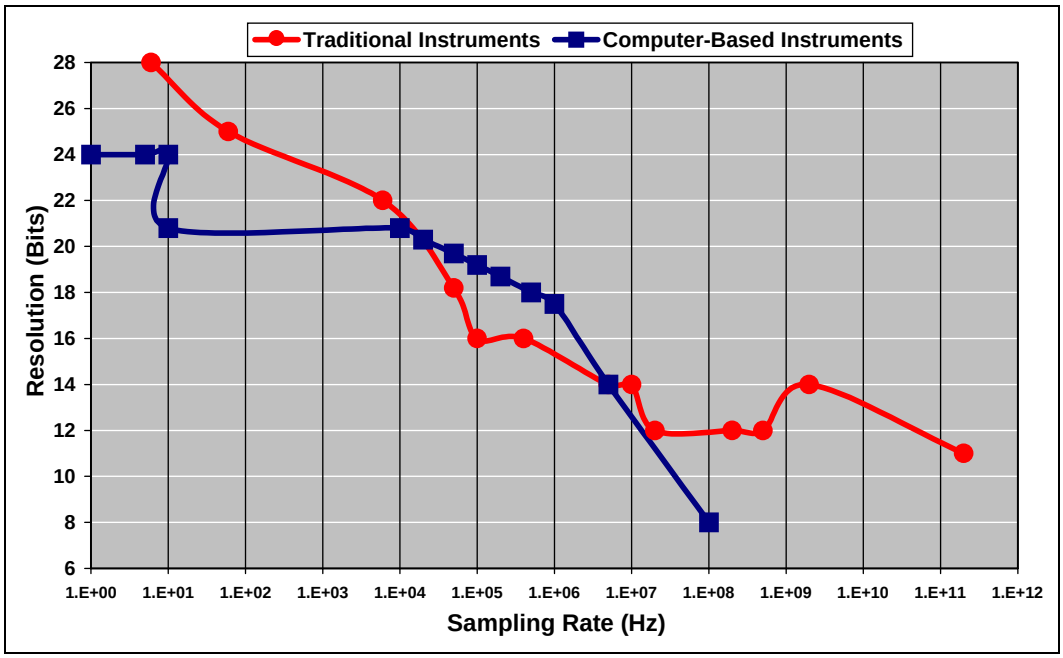



**Figure 4.** Digitizing Capabilities of Stand-Alone versus Computer-Based Instruments

Two key market forces drive innovation in the ADC market. First, semiconductor manufacturers armed with new process technologies can now cost-effectively produce high performance ADCs at a price that is viable to a wider audience. Secondly, large investments by telecom and consumer electronics companies motivate semiconductor manufacturers to invest in delivering these new, breakthrough ADC technologies. Driven by growing business and consumer demand for higher bandwidth communications, telecom corporations spend millions of dollars on high bandwidth wire-line and wireless technologies. The telecom industry needs faster flash ADC technology to push communication bandwidths to new levels (see Figure 5). Multimedia companies invest in high precision delta-sigma converters to meet the growing demands of the digital audio/video market for products such as DVD players, DTV equipment, and HDTV (see Figure 6).

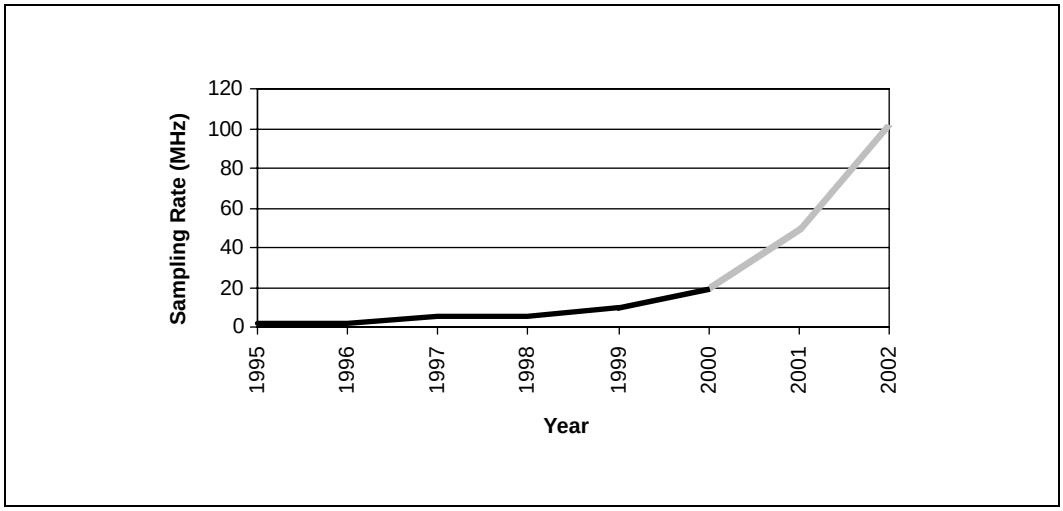



**Figure 5.** Projected 12-Bit ADC Technology Curve (Telecom)

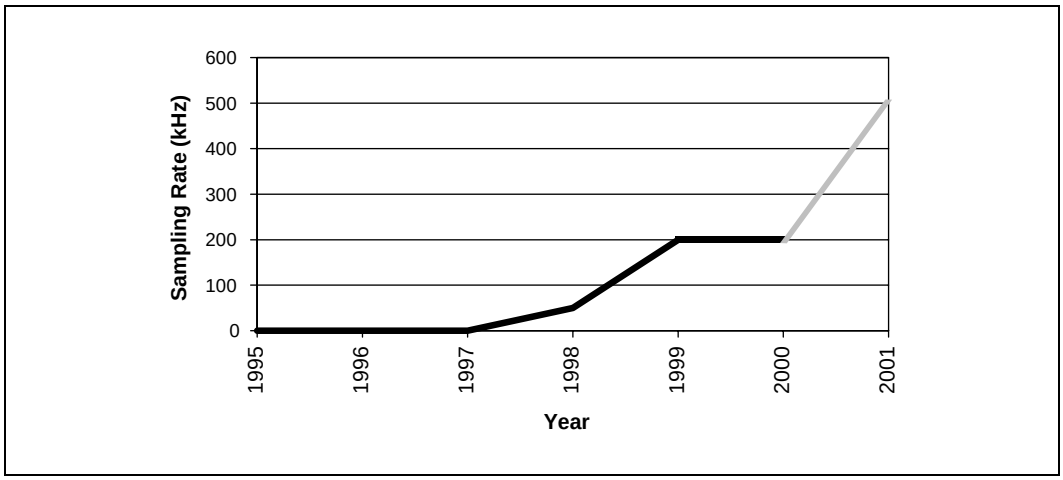



**Figure 6.** Projected 24-Bit ADC Technology Curve (Consumer Electronics)

**Note** The plot beyond 2000 in each graph denotes forecast projections. In addition, there were commercial off-the-shelf ADCs available before 1995, but these have been omitted because the initial cost made them infeasible.

# The Flexible Resolution ADC

Advances in ADC technology, combined with advances in reconfigurable logic, fueled innovation in the digital instrumentation paradigm. A new ADC technology developed by National Instruments, the Flexible Resolution ADC (Flex ADC), combines these two technologies. The Flex ADC used on the NI 5911 trades sampling for resolution (Figure 7) delivering 21-bit resolution at 10 kHz, 14 bits at 1 MHz, and 8-bits at 100 MHz. By using a 100 MHz 8-bit flash ADC, the Flex ADC achieves high precision at low sampling rates, maintaining at least 8-bits at 100 MHz without using expensive components. In addition, the range of possible measurements spans a wide spectrum of traditional instruments.

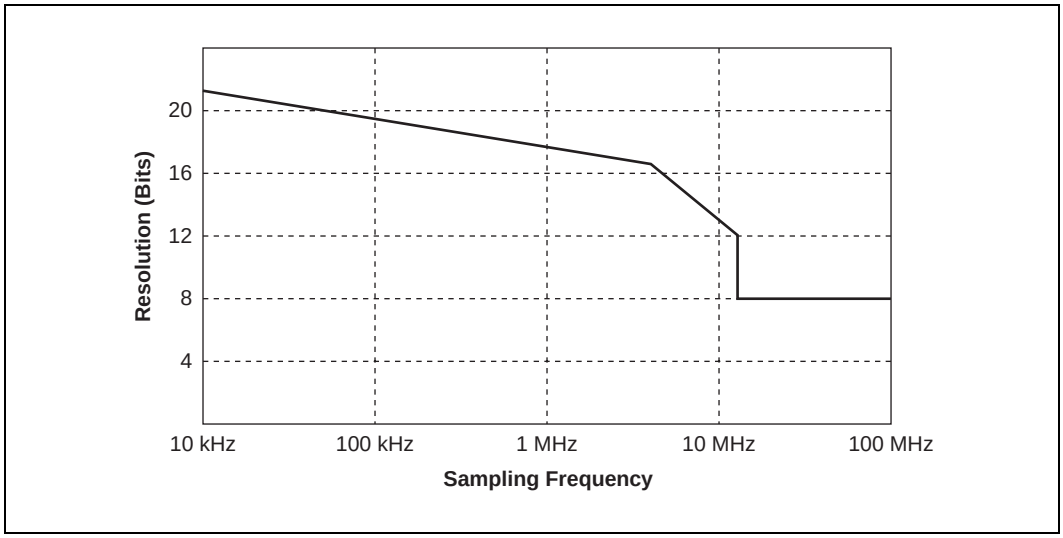



**Figure 7.** Flex ADC Measurement Capabilities (From data sheet)

To achieve its unique capabilities, the Flex ADC employs an 8-bit flash converter in a delta-sigma topology. Delta-sigma converters typically use high-speed 1-bit ADCs and achieve high resolution and dynamic range by decimating the acquired data. For example, to realize high dynamic range with a 1-bit ADC, the ADC typically samples at least 64 times the Nyquist frequency. Because 1-bit converters are inherently linear, linearity errors are minimized. But because the Flex ADC uses an 8-bit flash converter, it must take into account induced linearity errors. The Flex ADC compensates for the linearity errors by employing signal processing and noise shaping techniques in a Field Programmable Gate Array to filter out the noise and correct for the linearity errors. The amount of computational processing required to perform these techniques was not possible a few years ago. But because reconfigurable logic technologies have advanced in density and the cost per gate has consistently decreased, the Flex ADC delivers unprecedented performance while maintaining low cost. The range of measurement capabilities that the NI 5911 offers enables users to replace several instruments in a test system with a single NI 5911 and software.

# Of Digitizers and Instruments

Taking a “digitizer” view of the instrumentation world, you can compare stand-alone instruments across a continuum of sampling rate and resolution – the properties of the ADC technology that they employ. Figure 8 presents the mapping of instruments along these two attributes. DMMs and Data Loggers, which employ high precision ADCs, occupy the left side of the graph – low sampling rates but high resolution. Toward the right, frequency and dynamic signal analyzers appear with faster sampling rates, but lower resolution than DMMs. Communication testers need the high resolution (12-16 bits), but they also require faster sampling rates. Oscilloscopes occupy the next area under the curve – typically sporting eight-bit resolution. The last section on the graph, the far right, is reserved for spectrum analyzers, network analyzers, and vector network analyzers that need the highest sampling rates for measuring RF signals.

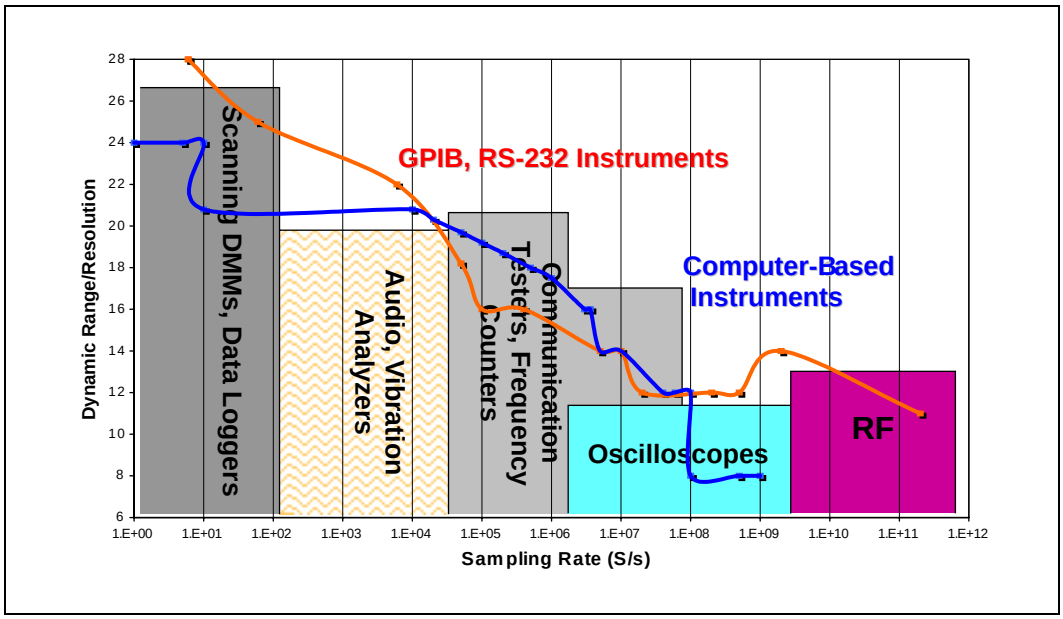



**Figure 8.** Traditional Instruments Viewed by ADC Capability

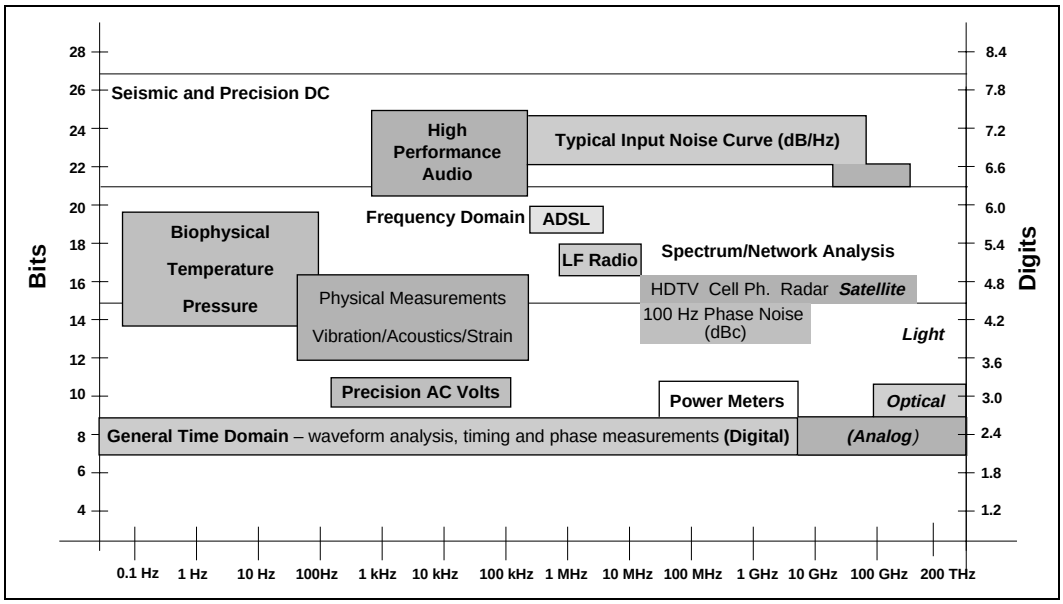



**Figure 9.** Common Measurements Curve

This point is further illustrated by application/measurement in Figure 9, which plots typical measurements by the sampling rate versus accuracy. Although this figure only displays a fraction of the vast universe of measurements, it generally shows where the driving applications fall.

# Stand-Alone Instrument Trends

Now that the ADC gap between stand-alone and computer-based instruments is narrowing, stand-alone instrument vendors differentiate their products in other ways in order to compete. Traditional *basic* instruments such as DMMs, oscilloscopes, and function generators cover common measurement ground within each instrument class (except for the extremes of the measurement spectrum). For more common measurements below the extremes, stand-alone instrument vendors add other differentiating features. Rather than decreasing the price, many times instrument vendors choose to keep their prices constant and justify the price by the value of the added features. As a result, users may purchase these basic instruments and all of their features whether they need them or not.

Another way to differentiate products is to solve more of the user's application by repackaging their generic "digitizer" technologies aimed at solving specific applications in a vertical market segment. Vertical markets, such as telecom and semiconductor, demand instruments with new capabilities to meet the challenges of their fast growing and continually changing environments. To capitalize on these opportunities, instrument vendors are developing application-specific instruments, which take a set of measurements aimed at a targeted market segment such as telecom. Although the digitizer technology used in these application-specific instruments may be the same as a standard product, instrument vendors develop and deploy custom software embedded inside the instrument to meet user needs. Application-specific instruments make it easy to solve a particular test challenge, but these instruments are typically much more expensive because the volumes are smaller and the user is charged a premium price.

## Put the PC inside the Instrument, or Put the Instrument inside the PC?

Stand-alone instrument vendors are integrating more of the PC and its capabilities inside the instrument to differentiate their products. By building more PC components inside their instruments, vendors effectively turn their instruments into “PCs.” What used to be closed box running a real-time or proprietary operating system has now become a PC, or a PC inside the box. By putting the PC inside the box, instrument vendors deliver some of the benefits of the PC to the user – the degree of benefit conveyed to the user is completely up to the instrument manufacturer. Moreover, instrument vendors can easily add new “PC-like” features to the instruments with minimal effort and cost, but the user usually ends up paying a price.

Alternatively, by placing the instrument inside the PC, the user has complete control over the PC, the application software, and the instruments that can be created. As technology evolves, the user immediately takes advantage of the advancement rather than waiting for the instrument vendor to offer it – with or without an additional fee. For instance, if a new networking technology such as gigabit Ethernet is developed, the user can simply purchase a new network interface board, install it in the PC, and improve the connectivity of the measurement system. In this example, users can create distributed measurement systems with incremental effort. As PCs continue to escalate in performance, users can simply buy a new PC – complete with the latest technology – and reuse the measurement hardware and software to increase the performance of the entire system.

Using the virtual instrument architecture of Digital Measurement Systems, users can choose low-cost hardware with raw acquisition or stimulus capabilities and build the rest in software. In addition, modular software technologies such as ActiveX and Interchangeable Virtual Instrument (IVI) drivers empower users to build virtual instruments without having to write the software from scratch. Integrated graphical development environments such as LabVIEW take full advantage of modular software components and empower users to transform the hardware into the instrument of choice (regardless of the hardware platform or bus interface). With the continued evolution in ADC technology, computer-based instruments and data acquisition boards continue to rival stand-alone instruments in the range of measurements that the user can make. Because Digital Measurement Systems take full advantage of the trends in ADC technology as well as PC technology, the component cost of the measurement system continues to decline. Not only can users save money by not paying for unnecessary features, but also they can enhance these lower cost measurement components with software to construct virtual instruments with unique capabilities.

## Digital Measurement Systems

Digital Measurement Systems encompass more than just a collection of instruments; they encompas the entire measurement system. Virtual Instrumentation defined the paradigm of software creating the capabilities of the instrument. In Distributed Virtual Instrumentation, the basic elements – acquisition, analysis, and presentation – can reside anywhere in the system or across a network. Digital Measurement Systems takes virtual instrumentation one step further to encompass a broader range of connectivity, control, software, and PC advances to enable users to build tightly integrated measurement systems that are open. Measurement components of these systems include computer-based instruments and data

acquisition boards, along with signal conditioning to add attenuation, filtering, and additional gain for low-level voltage measurements.

Users can also easily add image acquisition and motion control without having to purchase a separate stand-alone system. Integrating these components into a system requires the same effort as adding a new instrument. With high-speed connectivity technologies, users can distribute the elements of the measurement to one or a number of different computers across a network to build distributed systems never before thought possible. The measurement components that make up the Digital Measurement System are unified by common hardware architecture and compatible modular software components so that users can create powerful test, measurement, and automation systems for the lowest possible cost.

## The Revolution

The Measurement Revolution is about taking advantage of the advancements in PC and ADC technologies to build test, measurement and automation systems in new ways. PCs have realized performance gains of over 1,000 percent in the last 10 years – why haven't measurement systems? The answer is that the test and measurement industry is evolving. Just as digital instruments evolved from their analog predecessors, digital instruments will also evolve into virtual instruments. Virtual instrumentation systems do capitalize on the technology trends that are so prolific today. The next step in virtual instrumentation – Digital Measurement Systems – combines the benefits of innovation from different technological fronts to present a system approach where modular hardware and software components can be added easily to create low-cost, yet powerful systems that are flexible enough to solve today's test and measurement challenges. The results are truly revolutionary.

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